AI-ENABLED MORPHED IMAGE DETECTION AND DIGITAL FORENSIC ANALYSIS IN SECURE WEB SYSTEMS

A PROJECT REPORT

Submitted by

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in partial fulfillment of the requirements for the award degree of Bachelor of Engineering

20CS8504 PROJECT WORK

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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APRIL 2026

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**AI-ENABLED MORPHED IMAGE DETECTION AND DIGITAL FORENSIC ANALYSIS IN SECURE WEB SYSTEMS**" has been carried out by the students **SHARU RUBA S, UVASHINI NEKA B S, VARSHA VARDHINI R, VINITHA B.** The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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ABSTRACT

The rapid growth of digital media, social networks, and advanced image editing tools has significantly increased the risk of digital image manipulation and forgery. Morphed and forged images can easily mislead viewers and may even bypass biometric authentication systems such as facial recognition. These threats create serious concerns for digital security, online identity verification, and information reliability. AI-enabled system for detecting morphed and forged images using advanced image processing and digital forensic techniques. The system processes input images through several stages such as image acquisition, grayscale conversion, block extraction using Singular Value Decomposition (SVD), Gaussian Kernel Principal Component Analysis (PCA), and block matching to detect duplicated or altered regions. The proposed system improves the reliability of digital image verification and achieves an enhanced detection accuracy of 93%, making it suitable for secure web systems and digital forensic applications.

Keywords: Artificial Intelligence, Digital Image Forensics, Morphed Image Detection, Image Forgery Detection, Secure Web Systems, Image Processing.

ACKNOWLEDGEMENT

We thank our **Dr. N.Vasudevan**, Principal, for his valuable suggestions and support during the course of my research work.

We thank our **Mr.R.RajaVarman**, **M.E.,(Ph.D).,**Head of the Department, Assistant Professor (Sr.Grade), Department of Computer Science and Engineering, for his valuable suggestions and support for my research work.

We wish to record my deep sense of gratitude and profound thanks to my Guide **Ms.A.Dhivya Bharathi**, **M.E.,(Ph.D).,** Assistant Professor, Department of Computer Science and Engineering for her keen interest, inspiring guidance, encouragement with my work during all stages, to bring this thesis into fruition.

We are extremely indebted to our project coordinator **Mr.P.Matheswaran**, **M.E.,(Ph.D).,** Assistant Professor, Department of Computer Science and Engineering, for his valuable suggestions and support during the course of my research work.

We also thank the faculty and non-teaching staff members of the Computer Science and Engineering, K. Ramakrishnan College of Technology, Trichy, for their valuable support throughout the course of my research work.

Finally, we thank our parents, friends and our well wishes for their kind support.

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LIST OF SYMBOLS AND ABBREVIATIONS

CNN	-	Convolutional Neural Networks
AI	-	Artificial Intelligence
SIFT	-	Scale-Invariant Feature Transform
DCT	-	Discrete Cosine Transform
ANN	-	Artificial Neural Network
PCA	-	Principal Component Analysis
SURF	-	Speeded-Up Robust Features
GANS	-	Generative Adversarial Networks
GK-PCA	-	Gaussian Kernel Principal Component Analysis
VIT	-	Vision Transformers

CHAPTER 1
INTRODUCTION

1.1 DESCRIPTION

Digital images have become one of the most important sources of information in modern communication systems. Images are widely used in social media, journalism, security systems, government documents, and digital platforms to convey information and provide visual evidence. Because of their strong impact on people's perception, maintaining the authenticity of digital images has become extremely important.

With the advancement of powerful image editing tools and artificial intelligence technologies, it has become easier to manipulate digital images. Image manipulation techniques such as morphing, copy-move forgery, and feature alteration can create fake images that look realistic and difficult to detect with the human eye. These manipulated images can be used for spreading misinformation, committing identity fraud, or bypassing biometric security systems.

Image morphing is one of the most dangerous types of image manipulation. In morphing attacks, two or more facial images are blended together to create a synthetic image that resembles multiple individuals. Such images can deceive facial recognition systems used in passports, security systems, and identity verification processes.

The proposed system ai-enabled morphed image detection and digital forensic analysis in secure web systems is designed to detect manipulated or forged images using advanced image processing and digital forensic techniques.

1.2 DOMAIN SPECIFICATION

1.2.1 DIGITAL IMAGE FORENSICS DOMAIN

Digital image forensics is the primary domain of this project. It focuses on analyzing digital images to determine whether they have been manipulated or forged. Digital forensic techniques examine various characteristics of images such as pixel patterns, textures, and structural information.

In many cases, manipulated images appear visually identical to original images, making it difficult to detect forgery through manual inspection. Digital image forensics uses computational techniques to analyze hidden patterns and detect inconsistencies introduced during image manipulation.

This project applies digital forensic analysis to identify morphing attacks and copy-move forgeries. By analyzing image features and comparing image blocks, the system can detect manipulated regions and provide evidence of forgery.

1.2.2 IMAGE PROCESSING DOMAIN

Image processing plays an important role in the proposed system. It involves applying computational techniques to analyze and modify images for extracting meaningful information.

In this project, image processing techniques are used to perform preprocessing operations such as image acquisition, grayscale conversion, and block extraction. Converting the image into grayscale reduces complexity and allows faster processing while maintaining important visual details.

The image is divided into smaller blocks so that each region can be analyzed individually. These blocks are then processed to extract significant features that help detect suspicious or duplicated regions within the image. Image processing techniques enable the system to analyze images more effectively and prepare them for further forensic analysis.

1.2.3 ARTIFICIAL INTELLIGENCE DOMAIN

Artificial intelligence techniques are used in this project to improve the accuracy of forgery detection. Ai-based algorithms can analyze complex image patterns and identify hidden relationships within image data.

The proposed system uses gaussian kernel pca, which is an ai-based dimensionality reduction technique. It transforms image features into a higher-dimensional space where forged and original regions can be separated more effectively. Ai techniques allow the system to detect subtle differences in image structures that may not be visible to the human eye. This improves the reliability and accuracy of the digital forensic system.

1.3 Web Development Domain

Web development technologies are used to create a secure web-based interface for the proposed system. The system allows users to upload images through a web interface for forgery detection.

Frontend technologies such as JavaScript are used to build an interactive user interface that allows users to upload images and view results easily. The backend system processes the uploaded images and performs image analysis using forensic algorithms.

The web-based design allows the system to be easily accessed through digital platforms. This makes the system useful for organizations that need to verify image authenticity before accepting digital evidence or identity documents.

1.4 DATA ANALYSIS DOMAIN

1.4.1 DATA COLLECTION

The dataset used for this project consists of digital images that include both original and manipulated images. These images are collected from publicly available datasets and research resources related to image forgery detection.

The dataset contains various types of image manipulations such as morphing and copy-move forgery. These images are used to analyze how manipulated regions differ from original image structures.The collected dataset helps in testing and validating the effectiveness of the proposed detection system.

1.4.2 DATA PROCESSING

Before analyzing the images, preprocessing operations are performed to improve the efficiency of the system. Image preprocessing includes converting images into grayscale format and normalizing pixel values.

The grayscale conversion reduces the computational complexity while preserving essential image features. The processed images are then divided into overlapping blocks to capture local image characteristics. This step ensures that the system can analyze each region of the image effectively and detect suspicious patterns.

1.4.3 FEATURE EXTRACTION AND ANALYSIS

Feature extraction is an important step in detecting image forgery. In this project, singular value decomposition (svd) is applied to each image block to extract important feature vectors.

These feature vectors represent the structural characteristics of image blocks. The extracted features are then processed using gaussian kernel pca, which maps them into a higher-dimensional feature space.This transformation helps the system identify differences between original and manipulated regions. The feature vectors from different blocks are compared using block matching techniques to detect duplicated or altered regions.

1.5 SYSTEM INTEGRATION DOMAIN

The proposed system integrates multiple technologies including image processing, artificial intelligence, digital forensics, web development, and database management.

The system workflow includes image acquisition, grayscale conversion, block extraction using svd, feature transformation using gaussian kernel pca, and block matching for forgery detection. These modules work together to identify manipulated regions within the image.

A database system such as mysql is used to store image data and analysis results. The integration of these technologies creates a complete forensic framework capable of detecting morphed and forged images in secure web systems.

This integrated system improves digital security and helps prevent the misuse of manipulated images in various applications such as identity verification, digital evidence validation, and online security systems.

CHAPTER 2

LITERATURE SURVEY

1. DEEP LEARNING BASED IMAGE FORGERY DETECTION USING CONVOLUTIONAL NEURAL NETWORKS

Li Wei (2025) proposed a deep learning-based digital image forgery detection system using Convolutional Neural Networks (CNN). The primary objective of the research was to develop an automated system capable of detecting manipulated images such as copy-move forgeries, splicing attacks, and morphing manipulations. Traditional digital image forensic techniques rely on handcrafted feature extraction methods, which may fail when images undergo complex transformations or post-processing operations. Therefore, the study focused on implementing deep learning models that can automatically learn meaningful feature representations from large image datasets.

In the proposed system, the input image is processed through a convolutional neural network architecture that extracts hierarchical features from different layers of the image. CNN models can identify spatial patterns, texture inconsistencies, and structural distortions that indicate possible image manipulation. The network was trained using a large dataset containing both original and manipulated images. During training, the CNN model learns to differentiate between authentic and forged images by analyzing pixel-level and structural characteristics. Experimental results demonstrated that the CNN-based approach significantly improves detection accuracy compared to traditional block-based methods.

2. AI-BASED MORPHED FACE DETECTION USING TRANSFORMER MODELS

J. Kumar (2024) proposed an artificial intelligence-based morphed face detection system using transformer-based neural network architectures. Morphing attacks involve combining facial features from two different individuals to create a synthetic identity that resembles both persons. Such manipulated images can be used to bypass biometric authentication systems such as passport verification and facial recognition technologies. The research aimed to develop a robust system capable of detecting these morphing attacks by analyzing facial structures and texture patterns.The proposed model uses transformer-based deep learning techniques to analyze relationships between different regions of facial images. Unlike traditional convolutional neural networks that focus mainly on local features, transformer models capture long-range dependencies between image regions. This enables the system to detect subtle inconsistencies in facial landmarks, lighting patterns, and texture variations.The dataset used in the study consisted of genuine and morphed facial images generated using advanced morphing software tools.

The model was trained to detect inconsistencies by analyzing facial geometry, symmetry patterns, and pixel-level variations. Experimental results demonstrated that the transformer-based detection system achieved high accuracy in identifying morphed images.The research highlighted that transformer architectures provide improved feature representation and stronger detection capability for complex image manipulations. This approach is particularly effective in biometric security systems where reliable face authentication is required.

3. HYBRID IMAGE FORENSIC FRAMEWORK USING KERNEL PCA AND SVD FOR COPY-MOVE DETECTION

S. Patel (2023) proposed a hybrid digital image forensic framework that combines Kernel Principal Component Analysis (Kernel PCA) and Singular Value Decomposition (SVD) to detect copy-move forgery in digital images. The main objective of the study was to improve the accuracy and efficiency of block-based forgery detection techniques while reducing computational complexity.The proposed system begins by dividing the input image into overlapping blocks. This allows the algorithm to analyze local image regions where manipulation may have occurred. Singular Value Decomposition is applied to each block to extract singular values representing the structural information of that region. These singular values act as feature descriptors that capture important image patterns.Kernel PCA is then used to reduce the dimensionality of the extracted features while preserving the essential information required for detection. By projecting the features into a higher-dimensional space, Kernel PCA enhances the ability of the system to distinguish between original and duplicated regions within the image.Experimental results showed that the hybrid Kernel PCA–SVD framework significantly improves detection accuracy compared to traditional feature extraction techniques.

The method also reduces processing time by eliminating redundant information from the feature vectors.The study concluded that combining mathematical decomposition techniques with nonlinear dimensionality reduction methods provides an efficient solution for digital image forgery detection. To overcome these limitations, the proposed system utilizes CNN architecture, which can automatically learn hierarchical features from large datasets. The model processes images through multiple convolutional layers, extracting spatial and texture-based features. Pooling layers reduce dimensionality while preserving key information. During training, the system learns from labeled datasets containing both original and forged images.

4. SHRINKING THE SEMANTIC GAP: SPATIAL POOLING OF LOCAL MOMENT INVARIANTS FOR COPY-MOVE FORGERY DETECTION

Chao Wang (2022) proposed a novel copy-move forgery detection approach that focuses on reducing the semantic gap between low-level image features and high-level image interpretation. Traditional detection techniques rely mainly on pixel-level or texture-based features, which may not effectively capture complex patterns present in manipulated images.The proposed system utilizes local moment invariant features to represent the structural characteristics of image regions. Moment invariants describe the shape and orientation of image patterns and remain stable under geometric transformations such as rotation and scaling. These features are extracted from multiple regions of the image to capture the underlying structural information.Spatial pooling techniques are then used to aggregate the extracted features and capture spatial relationships between image regions. The pooled features are encoded using a bag-of-visual-words framework, which converts variable-length feature sets into fixed-length vectors suitable for comparison.Experimental evaluation demonstrated that the proposed method improves detection accuracy and robustness against common image manipulations such as compression, noise addition, and partial occlusion.

The research emphasized that using mid-level feature representations improves the ability of forgery detection systems to identify complex manipulation patterns, thereby enhancing the reliability of digital image forensic analysis. To overcome these limitations, the proposed system utilizes CNN architecture, which can automatically learn hierarchical features from large datasets. The model processes images through multiple convolutional layers, extracting spatial and texture-based features. Pooling layers reduce dimensionality while preserving key information. During training, the system learns from labeled datasets containing both original and forged images.

5. A ROBUST FORGERY DETECTION METHOD FOR COPY-MOVE AND SPLICING ATTACKS IN IMAGES

This research proposed by Manish (2023) a hybrid digital image forgery detection system capable of identifying both copy-move and image splicing attacks. Image splicing involves combining regions from different images to create a manipulated image, whereas copy-move forgery duplicates regions within the same image. Detecting both types of manipulation requires analyzing multiple image characteristics.The proposed system integrates Discrete Cosine Transform (DCT) and Local Binary Pattern (LBP) features to capture both frequency and texture information from the image. The detection process begins by dividing the input image into overlapping blocks. Each block is processed using DCT to extract frequency-domain information that reveals duplicated patterns.Local Binary Pattern is then applied to extract texture features representing the local structural characteristics of the image.

The combination of DCT and LBP features provides a comprehensive representation of the image content.These feature vectors are then classified using a Support Vector Machine classifier to determine whether the image contains manipulated regions. Experimental results showed that the hybrid approach achieves high detection accuracy even when images undergo compression, noise addition, or blurring.The study concluded that combining multiple feature extraction techniques improves the reliability and robustness of digital image forgery detection systems. To overcome these limitations, the proposed system utilizes CNN architecture, which can automatically learn hierarchical features from large datasets. The model processes images through multiple convolutional layers, extracting spatial and texture-based features. Pooling layers reduce dimensionality while preserving key information. During training, the system learns from labeled datasets containing both original and forged images.

6. COPY-MOVE FORGERY DETECTION USING SCALE INVARIANT FEATURES AND REDUCED LOCAL BINARY PATTERN HISTOGRAM

Jun young park (2020) proposed a copy-move forgery detection technique that integrates scale-invariant feature transform (sift) with reduced local binary pattern (rlbp) histogram features. The objective of this research was to improve the efficiency and accuracy of detecting duplicated image regions while reducing computational complexity. Traditional sift-based methods rely on high-dimensional descriptors that increase processing time and memory usage. Similarly, local binary pattern techniques generate large feature vectors that may contain redundant information. The proposed system addresses this limitation by introducing a reduced histogram representation that preserves essential texture features while minimizing computational cost.In the proposed framework, sift is first used to detect keypoints in the input image. These keypoints represent distinctive features that remain stable under transformations such as scaling, rotation, and illumination changes. Around each detected keypoint, a local image region is extracted and processed using the reduced local binary pattern operator.

This operator captures texture patterns within the image while reducing the dimensionality of the descriptor.The extracted features are then compared using similarity matching techniques to identify duplicated regions. Experimental results showed that the proposed sift–rlbp approach achieves high detection accuracy and reduces computational overhead compared to conventional sift-based methods.The research also demonstrated that the reduced descriptor representation improves scalability for high-resolution images and enables faster processing in large datasets. This makes the technique suitable for real-time digital image forensic applications. Through attention mechanisms, the system learns relationships between facial landmarks and identifies subtle irregularities.

7. COPY-MOVE FORGERY DETECTION USING SUPERPIXEL SEGMENTATION AND HELMERT TRANSFORMATION

This research introduced by ziau kong (2021) a copy-move forgery detection technique based on superpixel segmentation combined with helmert transformation. Traditional block-based detection methods divide images into fixed rectangular blocks, which may not accurately represent natural object boundaries. This limitation can reduce detection accuracy when manipulated regions have irregular shapes. The proposed approach addresses this issue by segmenting the image into superpixels that represent meaningful image regions.Superpixel segmentation groups pixels with similar color and texture characteristics, allowing the system to analyze image regions more effectively. After segmentation, the helmert transformation is applied to model geometric transformations such as translation, rotation, scaling, and reflection. This transformation enables the detection system to identify duplicated regions even when they have undergone geometric modifications.Feature descriptors are extracted from each superpixel and compared with descriptors from other regions of the image.

If similar patterns are detected between different superpixels, the algorithm identifies them as potential duplicated areas.Experimental results demonstrated that the superpixel-based approach improves boundary detection and reduces false positives compared to traditional block-based methods. The system also showed strong robustness against image manipulations such as compression and noise addition.The study concluded that region-based segmentation techniques can significantly enhance the effectiveness of digital image forgery detection by preserving important image structures and improving detection accuracy. Through attention mechanisms, the system learns relationships between facial landmarks and identifies subtle irregularities. Experimental results demonstrated high accuracy in detecting morphing attacks.

8. DETECTION OF COPY-MOVE IMAGE FORGERY USING SVD AND SEARCH ALGORITHM

Abhishek kashyap (2018) proposed a copy-move forgery detection method that integrates singular value decomposition (svd) with the cuckoo search optimization algorithm. The goal of the research was to improve the accuracy of detecting duplicated regions in manipulated digital images while optimizing algorithm parameters automatically.The proposed system begins by dividing the image into overlapping blocks. Singular value decomposition is then applied to each block to extract singular values representing the structural characteristics of the image. These singular values serve as feature descriptors that can be used to identify similar image regions.One of the major challenges in svd-based detection techniques is selecting appropriate parameters such as block size and similarity thresholds.

The study addresses this challenge by applying the cuckoo search algorithm, a nature-inspired optimization technique that automatically determines optimal parameter values.Experimental results demonstrated that the optimized svd approach significantly improves detection accuracy and reduces false matches. The system also showed strong robustness against common image manipulations such as compression, noise addition, and blurring.The research highlighted that combining mathematical decomposition techniques with optimization algorithms provides a more adaptive and efficient solution for digital image forgery detection systems. By projecting data into a higher-dimensional space, KPCA enhances the separation between original and forged regions. Experimental results showed improved detection accuracy and reduced computational complexity compared to traditional techniques. The study demonstrates that combining mathematical decomposition with nonlinear dimensionality reduction provides an efficient and reliable approach for forgery detection. This research introduced a region-based approach using superpixel segmentation and Helmert transformation.

9. THE ROBUST AND EFFICIENT COPY-MOVE FORGERY DETECTION TECHNIQUE BASED ON SIFT AND SVD

Rajeev rajkumar (2017) proposed a hybrid copy-move forgery detection technique that combines scale-invariant feature transform (sift) with singular value decomposition (svd). The objective of the research was to overcome the limitations of using either block-based or key point-based detection methods individually.The proposed system first applies the sift algorithm to detect key points within the image. These key points represent stable image features that remain consistent under geometric transformations such as rotation, scaling, and illumination changes. Around each detected key point, a small neighborhood region is extracted for further analysis.Singular value decomposition is applied to these regions to extract singular values representing structural information of the image. These values form compact feature vectors that are compared across different image regions to identify duplicated areas.

Experimental evaluation demonstrated that the hybrid sift–svd approach improves detection accuracy and reduces false positive results compared to conventional methods. The system also showed good performance when images were affected by noise, compression, and geometric transformations.The research concluded that integrating keypoint-based feature detection with mathematical decomposition techniques provides a robust solution for identifying manipulated regions in digital images. By projecting data into a higher-dimensional space, KPCA enhances the separation between original and forged regions. Experimental results showed improved detection accuracy and reduced computational complexity compared to traditional techniques. The study demonstrates that combining mathematical decomposition with nonlinear dimensionality reduction provides an efficient and reliable approach for forgery detection.

10. COMPARATIVE STUDY OF DIGITAL IMAGE FORGERY DETECTION TECHNIQUES

This research presented by jia Naug (2020) a comparative study of various digital image forgery detection techniques used in modern image forensic systems. The study analyzed several approaches including block-based detection methods, key point-based detection algorithms, and deep learning-based techniques. The objective of the research was to evaluate the strengths and limitations of these methods in detecting different types of image manipulation.Block-based approaches analyze images by dividing them into overlapping blocks and extracting statistical features such as discrete cosine transform, singular value decomposition, and principal component analysis. These methods are effective in detecting small duplicated regions but may require high computational resources.

Keypoint-based techniques such as scale-invariant feature transform and speeded-up robust features detect distinctive points in the image and compare their descriptors to identify duplicated areas. These methods are generally faster but may fail in low-texture regions.The study also examined deep learning techniques such as convolutional neural networks, which automatically learn complex feature representations from large datasets. Experimental results showed that hybrid approaches combining traditional feature extraction with machine learning techniques provide better performance.The research emphasized that selecting appropriate feature extraction and matching techniques is critical for improving the accuracy and reliability of digital image forgery detection systems. This study provides a comprehensive comparison of various forgery detection methods, including block-based, keypoint-based, and deep learning techniques. Block-based methods are effective but computationally intensive, while keypoint-based methods are faster but less accurate in low-texture regions.

CHAPTER 3

EXISTING SYSTEM

Digital image verification systems have been widely used in various applications such as biometric authentication, law enforcement, and digital media analysis. With the increasing use of digital images in identity verification systems and online communication platforms, ensuring the authenticity of digital images has become extremely important. Existing systems for detecting image manipulation primarily rely on traditional image processing techniques and manual forensic analysis methods.

Traditional digital image forensic systems focus on analyzing image metadata, compression artifacts, and pixel-level inconsistencies. Metadata analysis involves examining the information embedded within the image file, such as camera details, date of creation, editing history, and software used for image modification. While metadata can provide useful clues regarding image authenticity, it can also be easily modified or removed using simple editing tools. As a result, relying solely on metadata analysis is not considered a reliable method for detecting image manipulation.

Another approach used in existing systems is watermarking. In watermarking techniques, a hidden digital signature is embedded into an image when it is created. If the image is modified, the watermark may become distorted or lost, indicating that the image has been tampered with. Although watermarking provides a certain level of protection against image manipulation, it requires the watermark to be embedded during the image creation process. Many digital images available online do not contain embedded watermarks, which limits the effectiveness of this method.

Algorithms such as scale-invariant feature transform (SIFT) and speeded-up robust features (SURF) are commonly used to identify key points and detect similarities between different regions of the image. These techniques are particularly useful in detecting copy-move forgery, where a portion of an image is copied and pasted into another region of the same image.

Block-based forgery detection techniques are another approach used in existing systems. In this method, the image is divided into smaller overlapping blocks, and each block is analyzed separately. Feature extraction techniques such as discrete cosine transform (DCT) and principal component analysis (PCA) are applied to these blocks to identify duplicated patterns. By comparing features extracted from different blocks, the system can detect suspicious regions that may indicate image manipulation. Although these techniques can detect certain types of image manipulation, they have several limitations. One major limitation is their inability to detect advanced morphing attacks. Image morphing involves blending facial features from multiple individuals to create a composite image that resembles more than one person. Traditional image processing techniques often fail to detect such manipulations because the resulting image appears visually consistent. Another limitation of existing systems is their dependency on manual analysis. Many digital forensic tools require experts to carefully examine images and interpret the results generated by the system. This process is time-consuming and not suitable for applications that require real-time image verification.

Another significant challenge faced by existing systems is the increasing sophistication of image manipulation technologies. Advanced image editing software and artificial intelligence tools can generate highly realistic manipulated images that are extremely difficult to detect using traditional techniques.

Because of these limitations, there is a growing need for intelligent detection systems capable of identifying advanced image manipulation techniques. Modern research focuses on integrating artificial intelligence with digital image forensic analysis to improve detection accuracy.

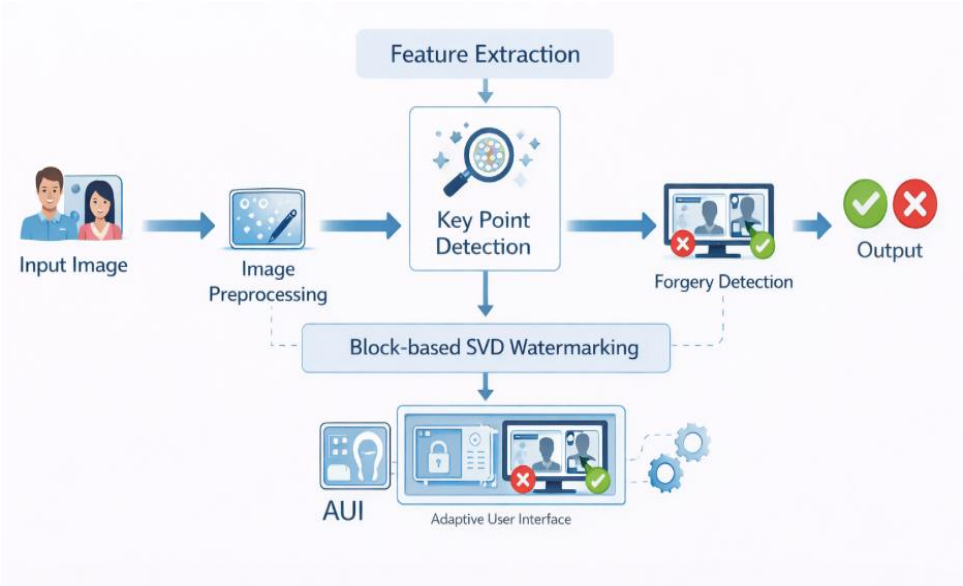


Figure 3.1: Existing System Architecture

In summary, existing digital image verification systems rely primarily on traditional image processing techniques and manual forensic analysis. Although these systems provide basic image verification capabilities, they are often unable to detect advanced morphing attacks and other sophisticated manipulation techniques.

AI-based detection systems use machine learning algorithms to analyze large datasets of authentic and manipulated images. Therefore, there is a need for more advanced systems that combine artificial intelligence, image processing, and web-based technologies to provide accurate and automated image verification.

CHAPTER 4

PROBLEM IDENTIFIED

4.1 RAPID GROWTH OF DIGITAL IMAGE MANIPULATION

The rapid advancement of digital technology and image editing software has made it very easy to manipulate images. Powerful editing tools allow users to modify images with high precision and realism. Because of this, forged images can be created in a way that looks almost identical to the original image. Many people use editing software to perform operations such as copy, paste, morphing, and object removal. These tools are widely available and require only basic technical knowledge. As a result, manipulated images can spread easily across social media and digital platforms.

Digital images are often used as evidence in many fields such as journalism, law enforcement, and scientific research. When these images are manipulated, it may lead to misinformation and incorrect conclusions. Therefore, verifying the authenticity of images has become very important. Traditional methods of identifying forged images rely on manual inspection. However, manual inspection is not reliable because human eyes cannot easily detect subtle changes in images. Advanced manipulation techniques make it even more difficult to identify tampering visually.

4.2 DIFFICULTY IN DETECTING COPY–MOVE FORGERY

Copy–move forgery is one of the most common types of image manipulation. In this technique, a portion of the image is copied and pasted into another region of the same image. This is often done to hide objects or duplicate elements within the image.Because of this similarity, detecting duplicated regions becomes very challenging. Many traditional algorithms fail to distinguish between original and copied regions.

Another challenge arises when the copied region is slightly modified before being pasted. Operations such as rotation, scaling, or blurring can be applied to the duplicated region. These modifications make the detection process even more complicated. When large images are analyzed, the system must examine many blocks within the image. Comparing all these blocks requires a significant amount of processing time and computational resources. This increases the complexity of the detection process.

4.3 HIGH COMPUTATIONAL TIME IN BLOCK MATCHING

Image forgery detection often involves dividing an image into many small blocks. Each block is analyzed and compared with other blocks to identify similar patterns. This process is known as block matching. In large or high-resolution images, the number of blocks becomes very high. As a result, the system must perform a large number of comparisons. This significantly increases the computational time required for analysis. Traditional techniques such as lexicographical sorting are commonly used for block comparison. However, these techniques are not always efficient for large datasets. They require extensive calculations and memory usage.

When computational time is high, the system becomes slow and inefficient. This makes it difficult to apply the method in real-time applications such as online image verification or surveillance systems. Therefore, reducing computational complexity while maintaining accurate detection is an important problem. An efficient approach is required to speed up the matching process without affecting performance.

4.4 LIMITATIONS OF EXISTING WATERMARKING METHODS

Digital watermarking is a technique used to embed hidden information into an image for security and authentication. Traditional watermarking methods usually embed the watermark into the entire image rather than specific regions. When the watermark is embedded globally, it may not effectively detect localized modifications. If only a small part of the image is tampered with, the system may fail to identify the exact forged region. Another limitation is that traditional watermarking techniques may affect the visual quality of the image. When the watermark is embedded strongly, it may distort the original image appearance.

Furthermore, some watermarking methods are not resistant to image processing operations such as compression, noise addition, or blurring. These operations may damage the watermark information and reduce detection accuracy. Because of these limitations, existing watermarking techniques are not always reliable for detecting advanced image manipulations. Improved watermarking methods are required to ensure better security and detection capability.

4.5 LACK OF RELIABLE VERIFICATION AND TRACKING MECHANISMS

In many digital systems, images are uploaded and shared through various platforms. However, most systems focus only on analyzing the image content without verifying the source of the image. Without proper verification mechanisms, it becomes difficult to track the origin of manipulated images. Attackers can upload forged images without revealing their identity, which makes investigation difficult. Information such as IP address, MAC address, and geographical location can provide valuable details about the uploader. However, this information is often not collected or used in traditional image forgery

CHAPTER 5

PROPOSED SYSTEM

The proposed system introduces an advanced framework for detecting morphed images and performing digital forensic analysis within a secure web-based environment. With the increasing use of digital images in authentication systems and online communication platforms, it is important to develop reliable techniques that can identify manipulated images. The proposed system integrates artificial intelligence, image processing, and digital forensic techniques to accurately detect image morphing attacks and other forms of digital image manipulation.

The primary objective of the proposed system is to develop an automated tool capable of verifying the authenticity of digital images uploaded through a web platform. The system analyzes the uploaded image using several processing stages including preprocessing, feature extraction, and classification. By examining the structural characteristics of the image, the system can identify anomalies that indicate possible manipulation.

In the first stage, the user uploads a digital image through a secure web interface. The web interface allows users to interact with the system easily and submit images for verification. Once the image is uploaded, it is stored securely in the system database for further analysis. The system ensures that user data is protected through secure authentication mechanisms and encrypted communication protocols. After the image is uploaded, the system performs image preprocessing. Image preprocessing involves preparing the image for analysis by improving its quality and removing noise.

Techniques such as image resizing, normalization, and filtering are applied during this stage. Preprocessing helps enhance important image features and improves the accuracy of subsequent analysis.

The next stage of the proposed system is featuring extraction. Feature extraction is a critical step in digital image forensic analysis. In this stage, important image characteristics such as edges, textures, color patterns, and pixel distributions are extracted from the image. These features represent the unique structure of the image and can be used to identify inconsistencies that may indicate manipulation.

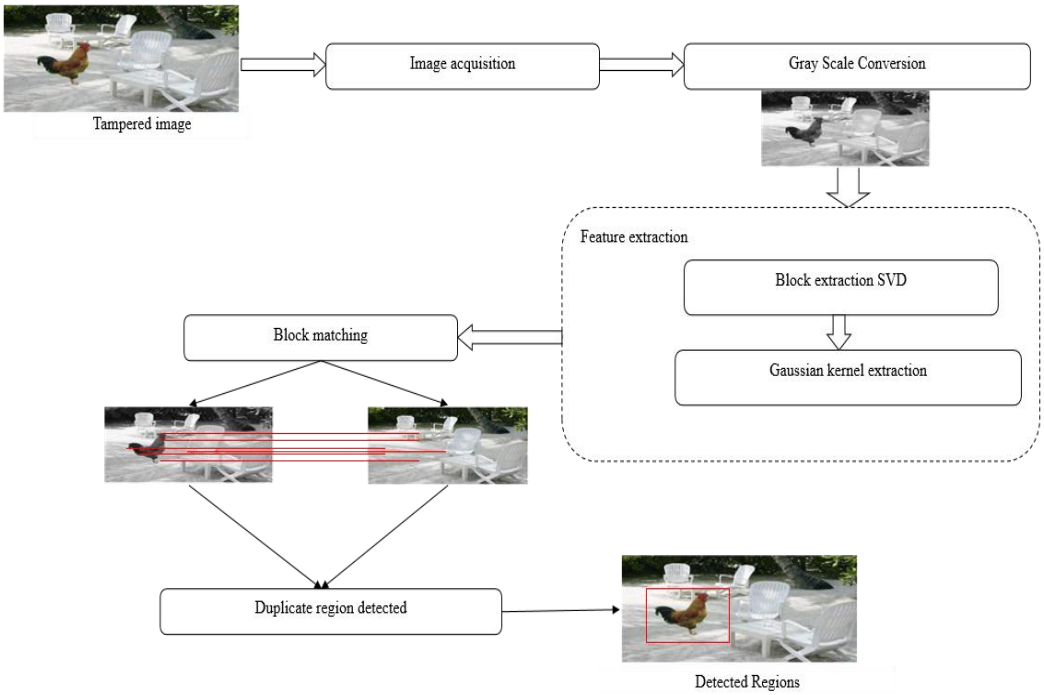


Figure.5.1: Proposed System

Machine learning algorithms are then applied to analyze the extracted features. The proposed system uses artificial intelligence models that have been trained using datasets of authentic and manipulated images.

Traditional detection techniques often fail to detect such manipulations because the resulting image appears visually realistic. The proposed system uses advanced feature analysis techniques to detect unnatural blending patterns and inconsistencies in facial structures. Another important feature of the proposed system is the use of block matching techniques to detect duplicated regions within an image.

In copy-move forgery, a part of the image is copied and pasted into another region to hide or alter information. Block matching algorithms divide the image into smaller sections and compare them to identify duplicated patterns. One of the key advantages of the proposed system is its ability to detect morphing attacks in facial images. Morphing attacks involve combining facial features from multiple individuals to create a composite image.

This helps the system detect manipulated regions accurately. When a new image is analyzed, the model compares its features with previously learned patterns to determine whether the image has been manipulated. Once the image analysis process is completed, the system generates a forensic report. The report indicates whether the image is authentic or manipulated. If manipulation is detected, the system highlights the suspicious regions within the image and provides a detailed analysis of the detected anomalies.

The proposed system also includes a secure web interface that allows users to access the verification results easily. The interface provides visualization tools that display the analyzed image and highlight detected manipulated areas. This helps users understand the results generated by the system. Compared to existing systems, the proposed system offers several advantages. It provides automated image verification, eliminating the need for manual forensic analysis. It also integrates artificial intelligence techniques that improve detection accuracy for complex manipulations such as morphing attacks.

Additionally, the web-based architecture allows the system to be accessed remotely, making it suitable for modern digital environments. The proposed system can be used in various applications including biometric authentication systems, passport verification systems, digital identity verification platforms, and cybercrime investigation processes. By detecting manipulated images effectively, the system helps maintain trust in digital media and improves the security of online platforms.

The proposed system ensures high accuracy by combining SVD, Gaussian Kernel PCA, block matching, and machine learning techniques. It reduces manual effort and provides automated, fast, and reliable image verification. Additionally, the system enhances security by tracking user activity using IP and MAC address logging, making it suitable for forensic and authentication applications.

In conclusion, the proposed system provides an intelligent and automated solution for detecting morphed images using digital forensic techniques and artificial intelligence. By combining image processing algorithms with machine learning models, the system can identify manipulated images with high accuracy and reliability.

CHAPTER 6

SYSTEM REQUIREMENT

6.1 HARDWARE REQUIREMENTS

• Processor

A computer system equipped with a minimum dual-core processor operating at 2.6 GHz is required for the effective functioning of the proposed system. The processor plays a vital role in executing Java programs, handling application logic, and managing system operations efficiently. Since the application involves multiple processes such as user interaction, data processing, and database communication, a reliable processor ensures smooth execution without delays. A higher-performance processor enhances system responsiveness, reduces execution time, and improves multitasking capabilities, thereby providing a better user experience.

• Memory (RAM)

The system requires a minimum of 1 GB RAM to execute the application efficiently. RAM is responsible for temporarily storing data and instructions that are actively used by the system. It enables the smooth functioning of the Java application, database connectivity, and background processes. Adequate memory ensures that the system does not slow down during execution and allows multiple operations to be performed simultaneously. Increasing the RAM capacity further improves system performance, especially when handling larger datasets or running multiple applications at once.

• Storage

A minimum of 160 GB hard disk space is required for storing application files, program components, database records, and user data. Storage plays a crucial role in maintaining system resources and ensuring data availability. It allows the system to store structured data in the MySQL database and retain application-related information for future use.

• Input Devices

Standard input devices such as a keyboard and mouse are required for interacting with the system. These devices allow users to enter data, provide commands, and navigate through the application interface. The keyboard is primarily used for typing inputs and commands, while the mouse helps in selecting options, clicking buttons, and interacting with graphical elements of the system. These input devices ensure efficient and user-friendly interaction with the application.

• Display Monitor

A 15-inch color monitor is required to display the graphical user interface of the system. The monitor enables users to view application outputs, interact with system features, and analyze results effectively. A good-quality display ensures clarity in visualization and enhances the overall usability of the system. The monitor plays an important role in presenting information in a structured and understandable format to the user.

6.2 SOFTWARE REQUIREMENTS

• Operating System

The proposed system is designed to run on Windows 10 Pro or higher. The operating system acts as a platform that supports the execution of application software and manages hardware resources. It provides an environment for running Java programs, database systems, and development tools. Windows operating systems are widely used due to their user-friendly interface.

• JavaScript

JavaScript is a high-level, interpreted programming language widely used for developing dynamic and interactive web applications. It is one of the core technologies of web development, along with HTML and CSS, and plays a crucial role in enhancing user experience by enabling real-time interaction within web pages.JavaScript is primarily used on the client side to create responsive and user-friendly interfaces.

• Java Programming Language

Java is also used as the backend technology to handle the core functionality of the system. It manages business logic, data processing, and communication between the user interface and the database. The backend ensures that user inputs are processed correctly and that data is stored and retrieved efficiently. Java provides strong support for multithreading, exception handling, and networking, which are essential for building reliable backend systems.

• MySQL

MySQL is used as the database management system for storing and managing system data. It is an open-source relational database that allows multiple users to access and manipulate data efficiently. MySQL ensures data consistency, integrity, and security. It supports structured query language (SQL) for performing operations such as data insertion, deletion, updating, and retrieval. The database plays a crucial role in maintaining records and ensuring smooth data flow within the system.

• Development Environment

NetBeans IDE is used as the development environment for implementing the project. It provides a comprehensive platform for writing, editing, debugging, and testing Java programs. NetBeans simplifies project management and improves development productivity.

CHAPTER 7

SYSTEM IMPLEMENTATION

7.1 IMAGE ACQUISITION

The rapid progress in colorization technologies has enabled colorized images to be visually indistinguishable from natural images. State-of-the-art colorization methods are already capable of misleading human observers in the subjective tests. Digital imaging or digital image acquisition is the creation of a digitally encoded representation of the visual characteristics of an object. Making an input of this created digital image is known as image acquisition. The image can be extracted with our features the digital sequences of our images. such as a physical scene or the interior structure of an object. According to our observation, the colorized images tend to possess less saturated colors, and the colorization method favors some colors over others, though these differences are difficult to visually detect. Since the Hue-Saturation-Value (HSV) color space separately represents the chrominance information in the hue and saturation channel, we calculate the normalized histograms (each containing 200 bins) of the hue and saturation channel in 15000 natural images and their corresponding fake colorized images, separately.

7.2 GRAY SCALE CONVERSION

The gray scale conversion is achieved since the computational values will be less. The gray scale conversion turns the image into black and white where the number of pixels will be very less. Here the pixel value will be 0-255 where they can be known as black and white. Instead of computing large values from the color images these computations between 0-255 reduce the complexity. A most recent work converted the color image and video to grayscale [8].

The proposed technique converted the image and video perceptually accurate. First, H-K (Helmoltz-Kohlrausch) phenomenon predicted by a chromatic lightness term that corrects perceived lightness based on the color's chromatic component. The color image converted to linear RGB by inverse gamma mapping, then transformed to CIELUV color space. Its apparent chromatic object lightness channel calculated. Lightness channel to grayscale values mapped using reference white chromatic values. Gamma mapping applied to move from a linear space to a gamma-corrected space. Local contrast increased in the grayscale image to represent better the local contrast of original image. The work carried out using CIELab and CIELuv color spaces. This two step approach a good compromise between a fully automatic technique (first step) and user control (second step) making this approach well suited for natural images, photographs, artistic reproductions as well as business graphics. The main limitation of the approach is the locality of the second step.

Algorithm

Step 1. Y = (0.299 x R) + (0.587 x G) + (0.114 x B);

Step 2. U = (B - Y) x 0.565;

Step 3. V = (R - Y) x 0.713;

Step 4. UV = U + V;

Step 5. R1=R*0.299;

Step 6. R2=R*0.587;

Step 7. R3=R*0.114;

Step 8. G1=G*0.299;

Step 9. G2=G*0.587;

Step 10. G3=G*0.114;

Step 11. B1=B*0.299;

Step 12. B2=B*0.587;

Step 13. B3=B*0.114;

Step 14. R4=(R1+R2+R3)/3;

Step 15. G4=(G1+G2+G3)/3;

Step 16. B4=(B1+B2+B3)/3;

Step 17. I1=(R4+G4+B4+UV)/4;

Step 18. END

The steps 1 to 3 calculate the luminance and chrominance values of the source color image. In the step 4 sum of chrominance value calculated. Steps 5 to 16 the RGB values are approximated using RGB components. Step 17 calculates the average of the four values R4, G4, B4 and UV. The I1 represents the resulted gray color image. The color image, 3D plot of the RGB values of the color image, and 3D plot of the reduced RGB values of the color image have shown and compared.

The results of the experiment carefully examined. The difference between the results of the proposed technique and recent techniques identified and discussed.

7.3 BLOCK EXTRACTION SVD

Image block matching is the main step of duplicated region detection for exploring copy-paste image forgery. For efficient block extraction Singular Value Decomposition (SVD) is imposed to make a block detection SVD requires data for all intervals corresponding to a time range query, and incremental SVD does not consider an arbitrary time range. SVD is a powerful numerical analysis tool for matrices computation.

It is a method for transforming a set of correlated variables into a set of uncorrelated variables. This property of SVD provides an interpretation of relationships among the original data items. The SVD is performed on the original image (F matrix)

$$F=UV^T \tag{5.1}$$

The original image is divided into blocks using the diagonal features extracted. Each object with the combination of the pixel are get blocked with the given SVDThe watermark (W matrix) is added to the SVs of the original matrix.

$$D=S+kW \tag{5.2}$$

The proposed SVD based watermarking scheme in which the watermark is added to the SVs of the whole image or a apart of it [3]. A single watermark is used in this scheme which may be lost due to attacks. segmented into blocks and the watermark is added to the SVs of each block in a modified manner.

7.4 GAUSSIAN KERNEL

In other words, the Gaussian kernel transforms the dot product in the infinite dimensional space. The infinite dimensional space is converted into a Gaussian function of the distance between points in the data space. If two points in the data space are nearby then the angle between the vectors that represent them in the kernel space will be small.

- For an image block (x, y) of size $h \times w$, where x, y are 0, 1, 2, . . . , $N - 1$, we decompose the block $Brc(x, y)$ in terms of 2D SVD basis function. The result occurs in the form of a coefficients matrix $C(p, q)$ of size $h \times w$ that contains the SVD coefficients:
- In our implementation, a Gaussian RBF kernel function is chosen, which is defined by the mapping function : $[0,\infty) \rightarrow$ R.
- The dimensionality of matrix MKPCA can be reduced to NC where π is the Gaussian kernel parameter.

- Using this parameter the PCA based difference are analyzed with the computed features.

However, this approach cannot provide fast query speed for a long time range due to heavy computations induced by the reconstruction and the following SVD. The goal of Stitched-SVD is to efficiently stitch the SVD results in the query time range by avoiding reconstruction and minimizing the numerical computation of matrix multiplication. Feature extraction reduces dimensionality with minimal loss of information by (linear or nonlinear) projection of D-dimensional vector onto d dimensional vector (d < D). n image analysis area, there are several types of feature extraction: frequency domain feature, transform-based feature, spatial domain feature, statistical feature, histogram and color feature, texture feature, edge feature. Each method has advantages and disadvantages.

Hence selecting the best feature is important and it depends on system specifications. In duplicated region detection to select and evaluate the feature extraction method. The similarity vector D = {d1, d2, · · · , dn−1} represents the sorted Euclidean distances regarding other feature descriptors. In the 2NN test, the ratio is calculated by dividing the closest distance by the second-closest distance; then, that value is compared with a threshold value

7.5 BLOCK MATCHING

In our implementation, the size of each block Bi is $h \times w$, where the value of h and w depends upon the bit block we provide. When the SVD is applied to block Bi, we get a coefficients matrix C of the same size that is further transformed through the Gaussian RBF kernel PCA. The feature vector with the block representation is identified with the diagonal representation extraction made by the SVD. The principal components are compared with 10 key points and then their variance is identified as the first principal component variance and second principal component variance.

In our process 10 most principal component values are selected for the block matching. The feature vector represents both the square and the rectangle block where the noise, blur and post processing technique variance is also noted. Unfortunately, we experimentally obtain an opposite result. In this approach the original matrix (F matrix) is divided into blocks and the watermark is embedded to the diagonal matrix (S matrix) of each block giving new matrices. An SVD is performed on each of these new matrices to get the SV matrices of the watermarked image blocks. Then, these SV matrices are used to build the watermarked image blocks. By combining these blocks again into one matrix of the original image dimensions, the watermarked image Fw is built in the spatial domain. The steps of embedding the watermark can be summarized

The first attack applied is Gaussian noise with zero mean and 0.01 variance. The second attack is blurring using a low pass filter of 3x3 window. The third attack is cropping half of the watermarked image. The fourth attack is JPEG compression. The fifth attack is rotation by 15 degree. The sixth attack is resizing from size 256×256 to 128×128 and then to 256×256. Figure (5) shows the extracted watermark and the correlation coefficient between each extracted watermark and the original watermark for the method of Liu. The results reveal that the value of the correlation coefficient is less than 50% for extracted watermarks under attacks except for the compression attack.

7.6 DUPLICATE REGION DETECTION

The target of the implemented system is to find duplicated regions where the similarity index between regions (feature vectors). If the similarity index is less than the overlapped value then the region is detected as forged region. Two conditions are imposed over the duplicated block detection procedure: (1) the blocks are nonintersecting and non-overlapping, and (2) the similarity index does not exceed a threshold. The Gaussian feature extraction with the SVD block overlapping helps to detect the forged area in the digital image with the threshold coordinates.

CHAPTER 8

SYSTEM TESTING

System Testing is the testing of a complete and fully integrated software product. Usually, software is only one component of a larger computer-based system that may include hardware devices, databases, networks, and other software modules. Therefore, the purpose of system testing is to verify that the entire system functions correctly as a whole. In this phase, the fully integrated system is tested to ensure that all components work together properly and that the system meets the specified functional and non-functional requirements.

8.1 UNIT TESTING

Unit testing is the process of testing individual modules or components of the software during the development stage. Each module is tested separately to verify that it functions correctly according to its design. Unit testing is usually performed by the programmer who develops the module. The main goal of unit testing is to detect coding errors and ensure that each function works as expected.

8.2 INTEGRATION TESTING

Integration testing is performed after unit testing to verify that different modules of the software work correctly when combined together. During this stage, multiple modules are integrated and tested to identify errors that may occur due to module interaction. Since large software systems often contain multiple modules developed by different programmers, integration testing ensures that all modules communicate correctly with each other.

8.3 VALIDATION TESTING

Validation testing is a quality assurance process used to verify whether the developed software meets the requirements and expectations of the stakeholders. It ensures that the system performs the functions that it was designed to perform. Validation testing checks whether the software satisfies the acceptance criteria defined during the requirement phase.

8.4 SYSTEM TESTING

System testing is conducted on the complete integrated system to verify that the entire application works correctly under different conditions. This testing stage ensures that the system meets functional requirements and operates correctly in a real-world environment. Professional testers usually perform system testing before the software is released to the market.

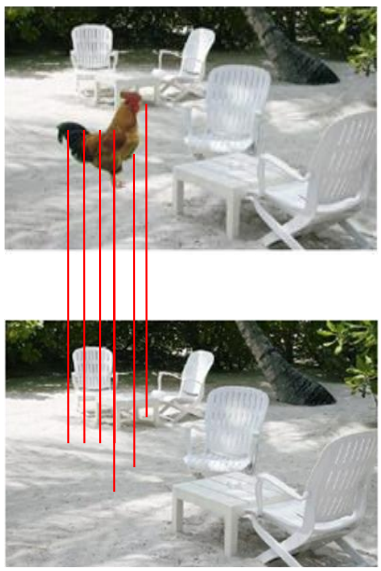
8.5 ACCEPTANCE TESTING

Acceptance testing is the final phase of testing performed before delivering the software to the end users. In this stage, the system is tested by actual users or clients to ensure that it satisfies their requirements. This process is often referred to as **beta testing**, where users evaluate the software and provide feedback. Acceptance testing confirms that the software is ready for deployment and meets the expectations of the end users.

CHAPTER 9

RESULTS AND DISCUSSION

In this section several experiments are carried out to compare between the methods of existing and the proposed approach. The 256x256 cock image is used to be watermarked. Figure 9.1 shows the original image, the watermark, the watermarked image, and the extracted watermark using Liu method. A single watermark is used. Figure 9.2 shows the original image, the block based watermark, the watermarked image and extracted watermark.



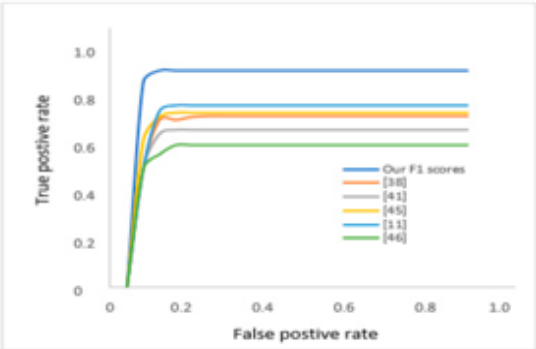


Figure 9.1: Represents the proposed system true positive blocks identification

The block of extracted watermarks which gives maximum correlation with the original watermark block is magnified in the figure. Similarly, Table (8.1) gives correlation coefficient results after applying lowpass filtering attacks with filters of different window sizes.

Correlation (1) refers to the maximum correlation obtained by the proposed method and correlation (2) refers to the correlation obtained by Liu method. These results reveal the ability of the proposed algorithm to extract watermarks even in the presence of severe attacks.

Method	Existing system	Proposed system
Variance	0.56	0.92
Correlation 1	0.61	0.72
Correlation 2	0.66	0.89
Number of blocks	10	19
Block identifies	2	9
Accuracy	89	97

Table 9.1: Represents the effects differ from proposed and the existing system

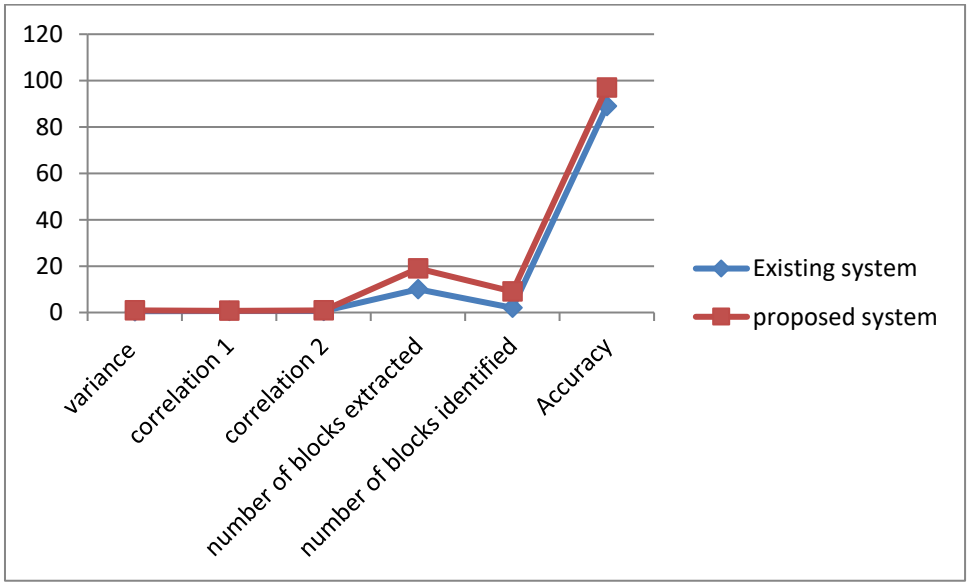


Figure 9.2: Comparison chart for the existing and proposed system using table

This implementation clearly proves that the proposed system makes a good accuracy compared to the existing system.

CHAPTER 10

CONCLUSION AND FUTURE WORK

10.1 CONCLUSION

This project presents a visually undetectable, robust watermarking scheme. The proposed algorithm depends on embedding the watermark into the SVs of the original image after dividing it into blocks. The experimental results show that the proposed Block-by-Block SVD-Based method gives fidelity and robustness against Gaussian noise, cropping and JPEG compression. One of the main problems in duplicated region detection is high computational time for exposing the similar blocks. In this paper we proposed a two layer block matching using two types of feature extraction namely low and high accurate features. Low accurate feature is used for grouping the blocks in first match while high accurate feature is applied in local block matching. In this method we did not focus on feature vector reduction for improving the time complexity. Due to reducing the feature vector dimension can affect robustness of the system; we used the block grouping for reducing the extra compare operations. As a result, this improves time complexity significantly. The results demonstrate that when using two layers matching, time complexity of the system is more efficient than lexicographically sorting algorithm.

10.2 FUTURE WORK

In future, we plan to extend our work to multilayer block matching with efficient multilayer feature extractions algorithms. The detection system will be extended to more transform domain watermarking approaches such as DWT-SVD and DCT-SVD.The system can be further improved by implementing multilayer feature extraction methods that capture both spatial and texture-based characteristics of facial images.

This approach will help in detecting high-quality morphing attacks that are difficult to identify using traditional methods.Another enhancement includes extending the system to support real-time morphing detection in authentication processes. This will allow faster identity verification in applications such as e-passports, security checkpoints, and online verification systems.

In addition, the Adaptive User Interface (AUI) can be refined to dynamically adjust based on user interaction and system feedback. This will improve usability, making the system more user-friendly while maintaining high detection performance.

Finally, future work can focus on increasing dataset diversity and improving model generalization. Training the system with large and varied datasets will enhance its ability to detect morphing attacks across different environments, lighting conditions, and image qualities.

APPENDIX-A

SOURCE CODE

code.java

```
package image morphing;

import static image_morphing. home. Title;

public admin_home

initComponents();

this.setTitle(title);

this.setLocationRelativeTo(null);

this.setResizable(false);

}

 @SuppressWarnings("unchecked")

private void initComponents()

 jPanel1 = new javax.swing.JPanel();

jLabel3 = new javax.swing.JLabel();

jButton1 = new javax.swing.JButton();

jButton3 = new javax.swing.JButton();
setDefaultCloseOperation(javax.swing.WindowConstants.DISPOSE_ON_CL
OSE);

 jLabel3.setFont(new java.awt.Font("Times New Roman", 1, 24)); // NOI18N

jLabel3.setText("ADMIN HOME");
```

```
jButton1.setText("View Data");

jButton1.addActionListener(new java.awt.event.ActionListener() {

public void actionPerformed(java.awt.event.ActionEventevt)
jButton1ActionPerformed(evt} })

jButton3.setText("Logout");

jButton3.addActionListener(new java.awt.event.ActionListener() {

public void actionPerformed(java.awt.event.ActionEvent evt) {‘

jButton3ActionPerformed(evt);

javax.swing.GroupLayout

jPanel1Layout = new javax.swing.GroupLayout(jPanel1);\

jPanel1.setLayout(jPanel1Layout);

jPanel1Layout.setHorizontalGroup(
jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEA
DING)

addGroup(jPanel1Layout.createSequentialGroup()

addGap(104, 104, 104)

addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alig
nment.LEADING)

addGroup(jPanel1Layout.createSequentialGroup()

addGap(47, 47, 47)

addComponent(jLabel3,
```

```
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alig
nment.LEADING, false)

addComponent(jButton1,

javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)

addComponent(jButton3, javax.swing.GroupLayout.PREFERRED_SIZE, 276,
javax.swing.GroupLayout.PREFERRED_SIZE

addContainerGap(113, Short.MAX_VALUE)));

getContentPane().setLayout(layout);

layout.setHorizontalGroup(
layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

addGroup(layout.createSequentialGroup()

addContainerGap()

javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,Short.MAX_VALUE)
layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

.addGroup(layout.createSequentialGroup()

addContainerGap()

addComponent(jPanel1,

javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)

}// </editor-fold>//GEN-END:initComponents
```

```
private void jButton3ActionPerformed(java.awt.event.ActionEvent evt)
{//GEN-FIRST:event_jButton3ActionPerformed

System.exit(0);

}//GEN-LAST:event_jButton3ActionPerform

private void jButton1ActionPerformed(java.awt.event.ActionEvent evt)
{//GEN-FIRST:event_jButton1ActionPerformed

new admin_view_data().setVisible(true);

}//GEN-LAST:event_jButton1ActionPerformed }
```

APPENDIX-B

SCREENSHOTS

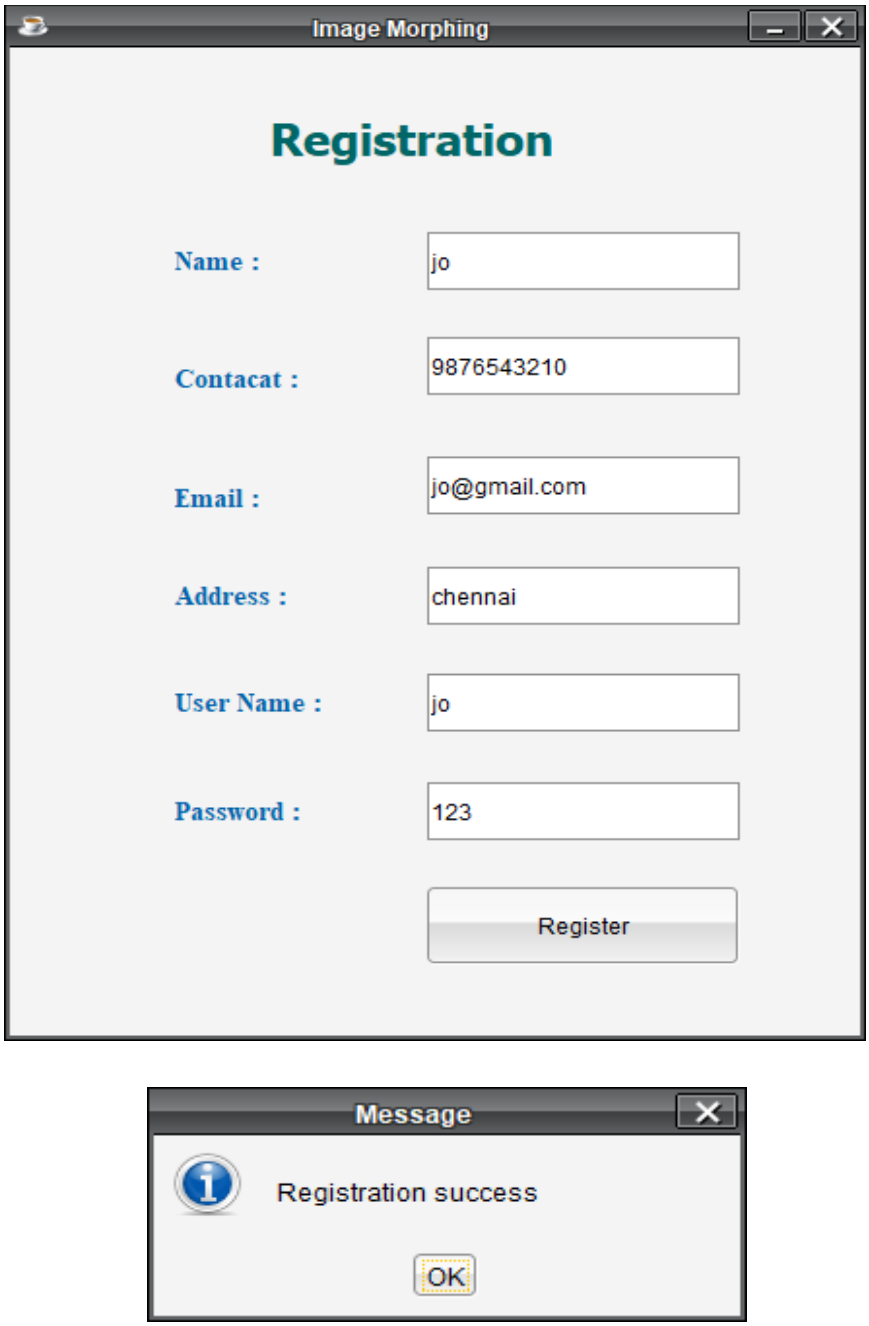


Figure B.1: Registration form

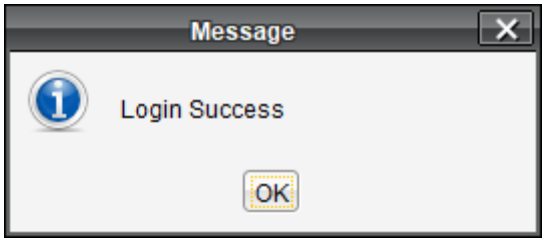


Figure B.2: User login system

Figure B.3: User home page

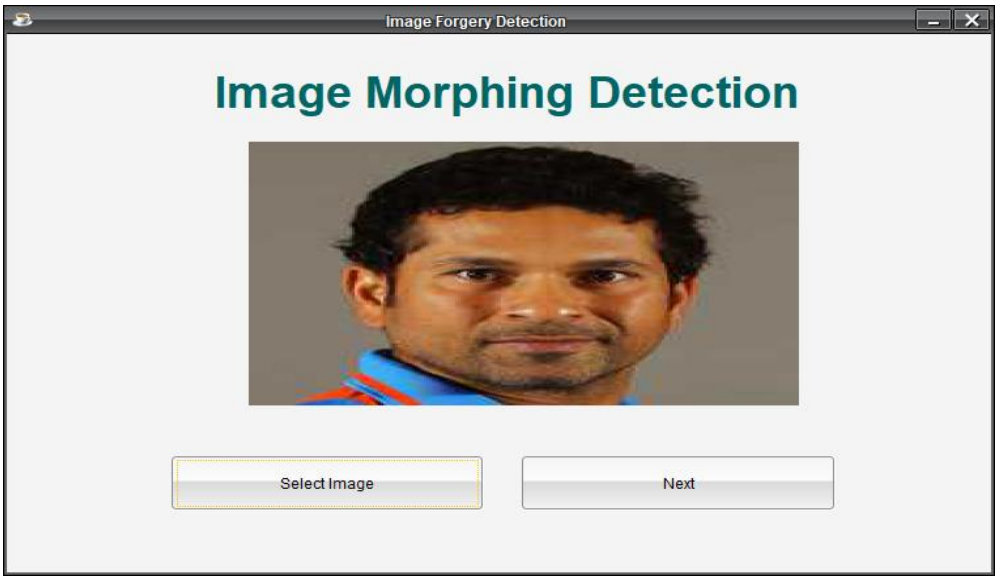


Figure B.4: image upload

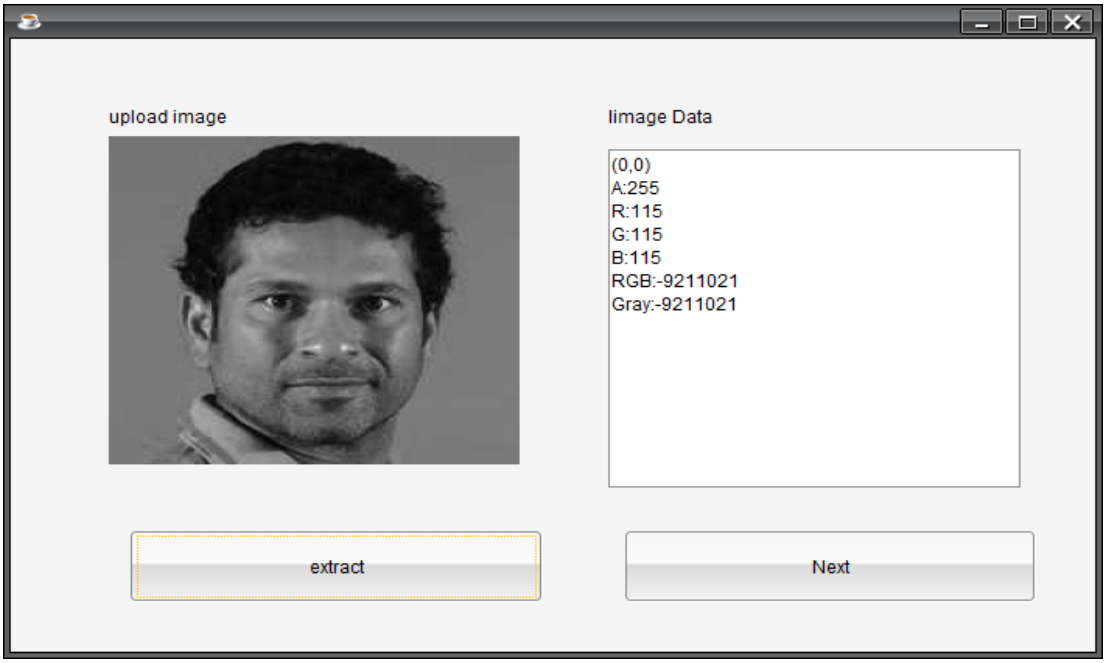


Figure B.5: Data Extraction

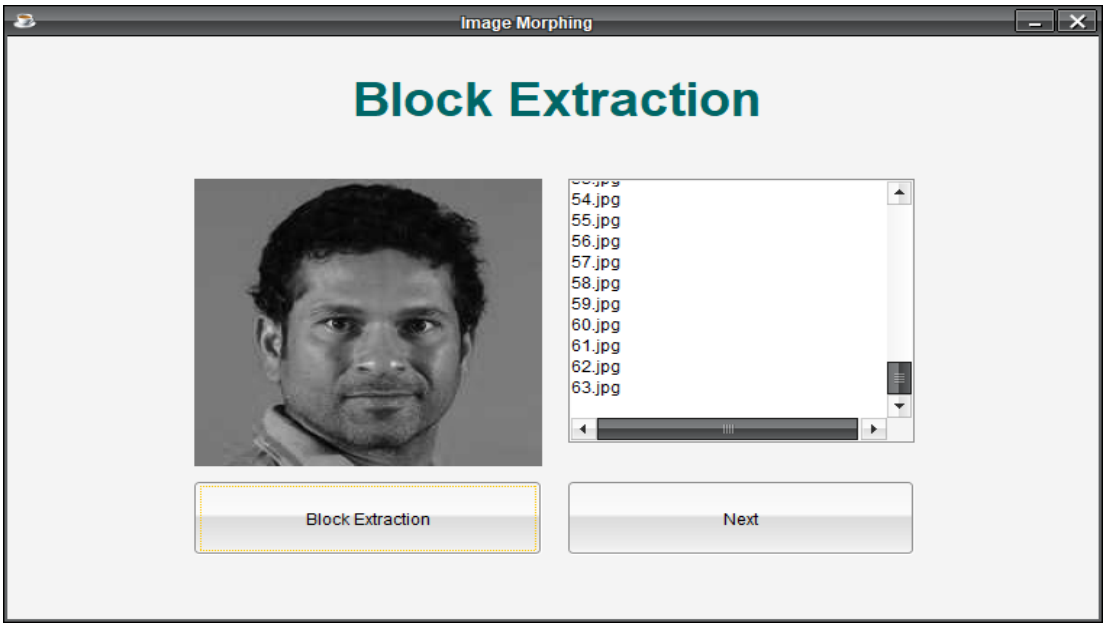


Figure B.6: Block Extraction

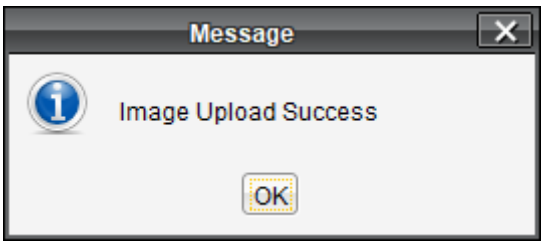


Figure B.7: Image Upload In Server

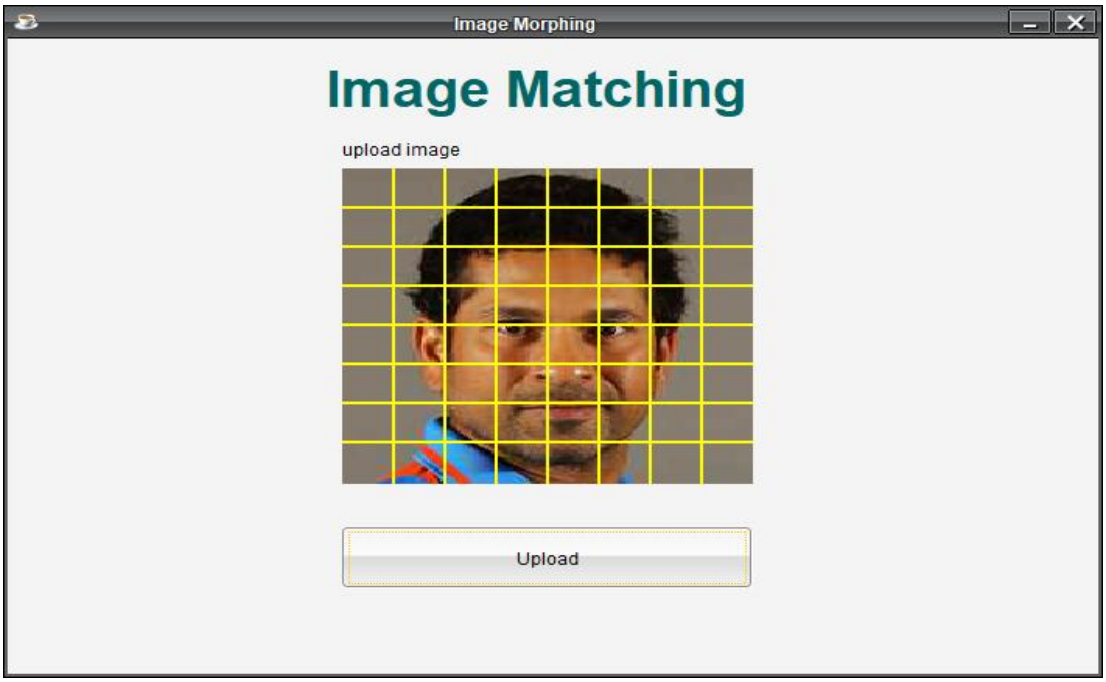


Figure B.8: Extracted blocks

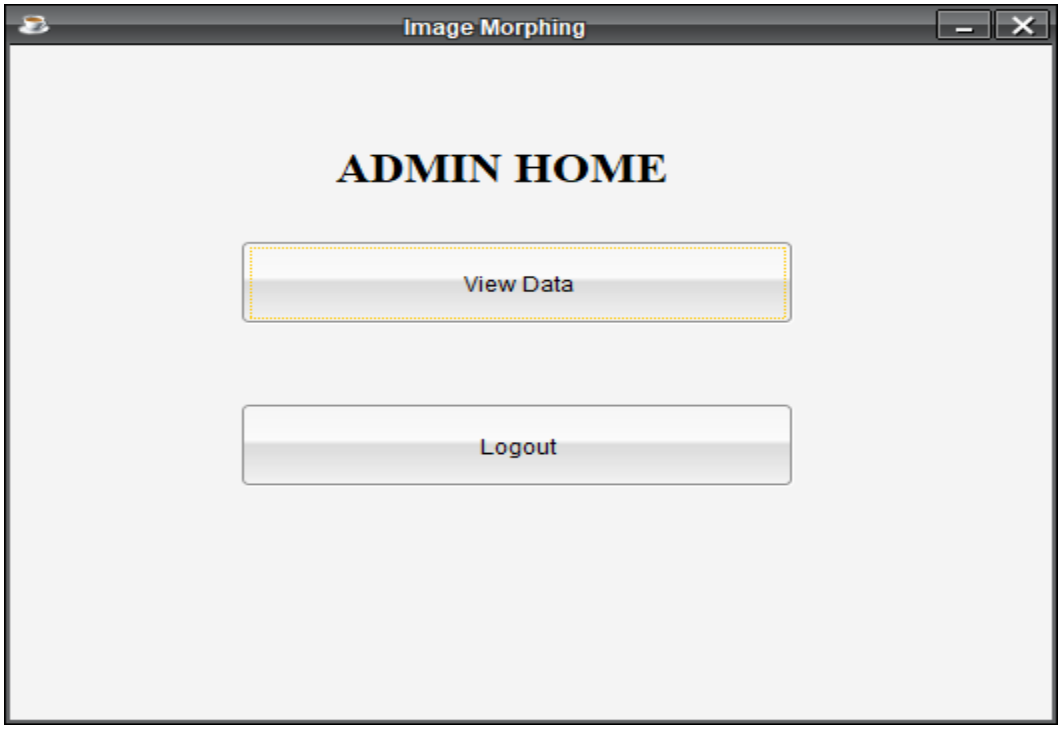


Figure B.9: Admin home page

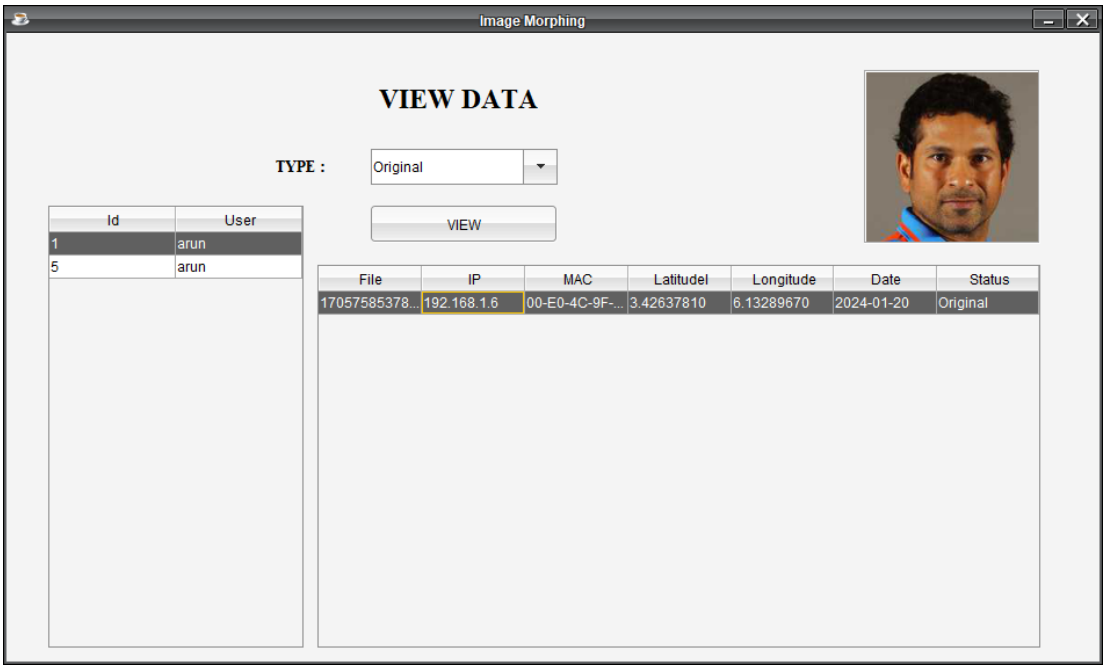


Figure B.10: Admin View Morphed Data

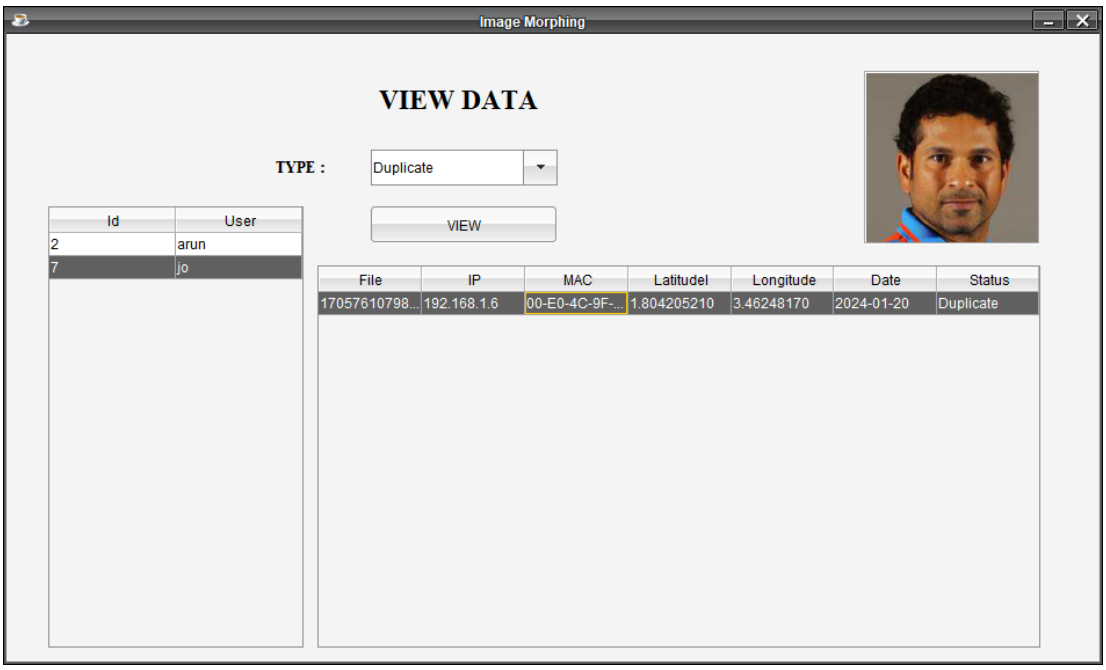


Figure B.11: Morphed Image User Details

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